

Fahima Islam

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#) | [ResearchGate](#)

RESEARCH INTERESTS

AI for Healthcare, Health Informatics, Medical Image Analysis, Computer Vision, Vision-Language Models, Large Language Models, Explainable AI, Deep Learning, and Trustworthy AI.

EDUCATION

Bachelor of Science in Computer Science and Engineering

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Jan 2023 – Sep 2026

CGPA: 3.95 / 4.00

Major in Information Systems: Introduction to Data Science; Data Warehouse and Data Mining; Computer Vision and Pattern Recognition; Machine Learning.

Thesis: *LiteTC-PolypNet++: A Semi-Supervised Boundary-Refined and Temporal-Consistency Guided Lightweight Framework for Stable Real-Time Polyp Segmentation.*

Higher Secondary Certificate

GPA: 5.00 / 5.00

2021

Secondary School Certificate

GPA: 5.00 / 5.00

2019

RESEARCH EXPERIENCE

Research Assistant

Elite Research Lab PLC, Queens, NY, USA (Remote)

Supervisor: Md Kishor Morol

April 2026 – Present

- Conducting a comprehensive audit of publicly available PCOS datasets (tabular and image) for data quality, leakage, duplication, and shortcut learning; identified criterion feature leakage in a high-visibility Kaggle dataset (23.8K downloads) and contributed to a verification atlas.
- Developing a multimodal deep learning pipeline for PCOS detection integrating ultrasound images with patient-matched clinical data, with a focus on robust fusion and real-world validation.

RESEARCH MANUSCRIPTS

Accepted and Presented / Accepted Papers

1. **Fahima Islam** et al. “The Follicle Count Paradox: Measuring Diagnostic Leakage in AI-Driven PCOS Prediction.” Accepted for presentation at the *4th International Conference on Computing Advancements (ICCA 2026)*, 2026. **[Accepted]**
2. **Fahima Islam** et al. “Improved Deep PACFNet: Multimodal Deep Learning for Cardiac Anomaly Detection Using Synchronized ECG-PCG Signals.” Presented at the *5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*, 2026. **[Accepted and Presented]**
3. **Fahima Islam** et al. “Enhancing Student Engagement Using Generative AI Tools in Online Learning Environments.” Presented at *ICETOL 2026*, Bremen, Germany, 2026. **[Accepted and Presented]**
4. **Fahima Islam** et al. “NeuroEffiNet: Efficient Deep Transfer Learning for MRI-Based Brain Tumor Diagnosis.” Accepted for oral presentation at the *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*, 2026. **[Accepted]**

Under Review

- **Fahima Islam** et al. “DeepLabNet: Lightweight Liver Ultrasound Segmentation via Haar Frequency Fusion and Entropy Guided Boundary Refinement.” Submitted to the *2026 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES 2026)*, 2026. **[Under Review]**
- **Fahima Islam** et al. “Conflict Aware Trust-Based Aggregation for Poisoning Resistant Federated Medical Image Classification.” Submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*, 2026. **[Under Review]**
- **Fahima Islam** et al. “Pressure Ulcer Classification Using Strong Segmentation-Guided Feature Fusion and DINOv2.” Submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*, 2026. **[Under Review]**
- **Fahima Islam** et al. “CATS-Thyroid: A Confidence-Aware Two-Stage Machine Learning Framework for Thyroid Disease Screening and Subtype Classification.” Submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*, 2026. **[Under Review]**
- **Fahima Islam** et al. “SunLite: A Sub-Megaparameter Ordinal Classifier for Sunflower Growth-Stage and Health Recognition.” Submitted to *NSysS 2026*, 2026. **[Under Review]**
- **Fahima Islam** et al. “X-RetinaPM: Explainable Deep Retinal Representations Driven by Lesion Supervision for Pathologic Myopia Analysis.” Submitted to the *3rd IEEE International Conference on Computing, Applications and Systems*, 2026. **[Under Review]**

- **Fahima Islam** et al. “CerebriLite-CAM: A Lightweight Interpretable Attention Network for Brain Tumor Classification from MRI.” Submitted to the *3rd IEEE International Conference on Computing, Applications and Systems*, 2026. **[Under Review]**
- **Fahima Islam** et al. “A Review on Machine Learning Approaches for Early Detection of Autosomal Dominant Polycystic Kidney Disease.” Review manuscript in preparation, 2026. **[In Preparation]**

TECHNICAL SKILLS

Programming Languages:	Python, Java, C, C++, C#, R, PHP, JavaScript
Web Technologies:	HTML5, CSS3, JavaScript, PHP
Databases:	MySQL, Oracle Database, Microsoft SQL Server, phpMyAdmin
Machine Learning / AI:	Machine Learning, Deep Learning, Regression, Classification, NLP, Computer Vision, Medical Image Analysis, Data Analysis
Development Tools:	Git, GitHub, XAMPP, Cisco Packet Tracer, EMU8086
Analytics / Scientific Tools:	RStudio, MATLAB
Design / Collaboration:	Figma, Draw.io, ClickUp, AutoCAD, OpenGL, GLUT
Productivity Tools:	Microsoft Word, Excel, PowerPoint

PROJECTS

- **Online Hotel Management System** — Developed a web-based system using HTML, CSS, JavaScript, PHP, phpMyAdmin, and XAMPP with admin login, room management, customer records, housekeeping, booking, payment, and billing modules.
- **Mental Health Outcome Prediction from Lifestyle Data** — Built regression and classification models in R to analyze mental-health outcomes from lifestyle-behavior datasets.
- **Automated NLP Pipeline on Amazon Food Reviews** — Implemented sentiment analysis, topic modeling, and clustering using TF-IDF, Naïve Bayes, LDA, PCA, and K-Means.
- **Urban-Grow: Smart Gardening for City Residents** — Designed a smart-gardening software engineering project with UML diagrams, SDLC documentation, community forums, marketplace, payment flow, and dashboard modules.
- **BuyNix Online Shop Management System** — Developed an e-commerce management system using .NET Framework, C#, and Microsoft SQL Server for product, order, and user management.
- **Tech Greek Online Gadget Shop** — Built a Java-based gadget shop system with inventory and customer-management features.
- **Waffle Verse Restaurant Management System** — Implemented a Java-based restaurant management application for order processing, billing, and menu services.
- **Game Management System** — Created an Oracle Database system for managing game records and user data.

LEADERSHIP AND SERVICE

Assistant General Secretary, Public Relations, AIUB Computer Club	2023 – 2026
Campus Ambassador, Augmedix Bangladesh	2024 – 2026
Local Volunteer, NASA International Space Apps Challenge	2025
Volunteer, Microsoft Office Specialist Bangladesh Championship	2025
General Member, Team Lifeline	2024 – 2026
News Reporter, Nirvik TV	2021 – 2025
Upazila Leader, Shorno Kishoree	2018 – 2023
Volunteer, JAAGO Foundation	2018 – 2025
Volunteer, UNICEF Bangladesh	2019 – 2021

TEACHING, TRAINING AND CERTIFICATIONS

IT Essentials Course, Cisco Networking Academy

- Completed foundational training in computer hardware, software, networking, troubleshooting, and IT support.

HONORS AND AWARDS

Merit-Based Scholarship to Pursue BSc in CSE, up to 60%	
Dean’s List Award, American International University-Bangladesh	6 times

LANGUAGES

English: Fluent

REFERENCES

Prof. Dr. Dip Nandi Professor, Dean American International University-Bangladesh dip.nandi@aiub.edu	Dr. Afroza Nahar Professor American International University-Bangladesh afroza@aiub.edu	Dr. Abhijit Bhowmik Associate Professor American International University-Bangladesh abhijit@aiub.edu
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